



KENYA INSTITUTE OF CURRICULUM DEVELOPMENT
Nurturing Every Learner's Potential

SENIOR SCHOOL CURRICULUM DESIGN

GRADE 10

CHEMISTRY

MARCH, 2025
First Published in 2025

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ISBN: 978-9914-52-913-5

Published and printed by Kenya Institute of Curriculum Development

FOREWORD

The Government of Kenya is committed to ensuring that policy objectives for Education, Training and Research meet the provisions of the Kenya Constitution 2010 and Vision 2030, East African Community (EAC) Protocol, African Agenda 2063, United Nations Sustainable Development Goals (SDGs) and demands of the 21st Century.

Towards achieving the vision and mission of Basic Education, the Ministry of Education (MoE) has successfully and progressively rolled out the Competency Based Curriculum (CBC) at Pre-Primary (PP1& PP2), Primary (Grades 1-6) and Junior School (Grades 7-9) levels of education. The first cohort of CBC learners will transit to Senior School (Grades 10-12) in 2026. The curriculum designs at this level facilitate learners to acquire relevant knowledge, skills, attitudes and values in preparation for Tertiary Education and the world of work.

The Grade 10 curriculum designs, like the designs for earlier levels, are anchored on the Kenyan National Goals of Education. They present essence statements, general and specific expected learning outcomes as well as strands and sub strands for the learning areas. The designs also outline suggested learning experiences, key inquiry questions, core competencies, Pertinent and Contemporary Issues (PCIs), values and suggested assessment.

I urge all Government agencies and other stakeholders in Education to use the designs to plan for effective and efficient implementation of the Competency Based Curriculum.

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PREFACE

The Ministry of Education (MoE) through the State Department for Basic Education, has been implementing the Competency Based Curriculum (CBC) since the adoption of the *Basic Education Curriculum Framework* (KICD, 2017) during the national convention in January 2017. The CBC was piloted in 2017 and 2018 and rolled out at Pre-Primary, Primary and Grades 1 to 3 in 2019. The curriculum was then introduced in the subsequent grades through a phase in - phase out strategy, to ensure a smooth transition. Currently, the first Cohort of CBC are at Grade 9 – the last grade at Junior School. The implementation of CBC has also benefited immensely from the recommendations of the Presidential Working Party on Education Reforms (PWPER) that were released in August 2023 and fully implemented, in terms of Curriculum design and curriculum support materials.

Grade 10 is the entry class of the Senior School (SS) in Basic Education, a very critical level where learners shall have the opportunity to focus on pathways and tracks of their choice, guided by their potentials and interests as demonstrated at earlier levels. The Senior School shall; therefore, form a foundation for further education and/or development of initial competencies for the Industry. This is an important step in the realisation of the Vision of the curriculum reforms; which is to produce *engaged, empowered and ethical citizens*. The vision and mission of CBC as enshrined in the Sessional Paper No. I of 2019 whose title is: *Towards Realizing Quality, Relevant and Inclusive Education and Training for Sustainable Development* in Kenya.

The Grade 10 curriculum designs indicate, among others, suggested learning experiences and resources. The Ministry of Education, through the State Department for Basic Education commits to provide these resources and relevant guidelines for a smooth implementation of the CBC. It is expected that the designs will guide teachers and other officers involved in supporting and quality assuring curriculum implementation to effectively facilitate learners to attain the expected learning outcomes for Grade 10 specifically and Senior School in general.

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ACKNOWLEDGEMENT

The Kenya Institute of Curriculum Development (KICD) Act Number 4 of 2013 mandates the Institute to develop curricula and curriculum support materials for Basic and Tertiary Education. The curriculum development process involves thorough research, and robust stakeholder engagement. Through a systematic and consultative process, the KICD conceptualised the Competency Based Curriculum (CBC) as captured in the *Basic Education Curriculum Framework* (KICD, 2017).

KICD receives its funding from the Government of Kenya (GOK) to enable the successful achievement of its mandate. The

Institute also receives support from development partners targeting specific programmes. The Grade 10 curriculum designs were developed with the support of the World Bank through the Kenya Primary Education Equity in Learning Programme (KPEELP); a project coordinated by MoE. Therefore, the Institute is very grateful to the GOK, through the MoE and the development partners for support in terms of policy guidance, resource provision and capacity building. Specifically, I wish to express special thanks to the Cabinet Secretary-MoE and the Principal Secretary-State Department for Basic Education.

We also acknowledge the KICD staff, all teachers, educators who took part as panelists and various stakeholders. In relation to this, we acknowledge the support of the Chief Executive Officers of the Teachers Service Commission (TSC), the Kenya National Examinations Council (KNEC) and other agencies in the sector for their support in the process of developing these designs.

Finally, we are very grateful to the KICD Council for consistent guidance and oversight. We assure all teachers, parents and other stakeholders that these curriculum designs will effectively guide implementation of the CBC at Grade 10 and preparation of learners for transition to further education and training or the world of work.

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NATIONAL GOALS OF EDUCATION

Education in Kenya should:

1. Foster nationalism and patriotism and promote national unity.

Kenya's people belong to different communities, races and religions, but these differences need not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationhood by resolving conflicts and promoting positive attitudes of mutual respect, which enable them to live together in harmony and foster patriotism in order to make a positive contribution to the life of the nation.

2. Promote the social, economic, technological and industrial needs for national development.

Education should prepare the youth of the country to play an effective and productive role in the life of the nation. **a) Social Needs**

Education in Kenya must prepare children for changes in attitudes and relationships which are necessary for the smooth progress of a rapidly developing modern economy. There is bound to be a silent social revolution in the wake of rapid modernisation. Education should assist our youth to adapt to this change.

b) Economic Needs

Education in Kenya should produce citizens with the skills, knowledge, expertise and personal qualities that are required to support a growing economy. Kenya is building a modern and independent economy, which is in need of an adequate and relevant domestic workforce.

c) Technological and Industrial Needs

Education in Kenya should provide learners with the necessary skills and attitudes for industrial development. Kenya recognises the rapid industrial and technological changes taking place, especially in the developed world. We can only be part of this development if our education system is deliberately focused on the knowledge, skills and attitudes that will prepare our young people for these changing global trends.

3. Promote individual development and self-fulfilment

Education should provide opportunities for the fullest development of individual talents and personality. It should help children to develop their potential interests and abilities. A vital aspect of individual development is the building of character. **4. Promote sound moral and religious values.**

Education should provide for the development of knowledge, skills and attitudes that will enhance the acquisition of sound moral

values and help children to grow up into self-disciplined, self-reliant and integrated citizens.

5. Promote social equity and responsibility.

Education should promote social equality and foster a sense of social responsibility within an education system which provides equal educational opportunities for all. It should give all children varied and challenging opportunities for collective activities and corporate social service irrespective of gender, ability or geographical environment.

6. Promote respect for and development of Kenya's rich and varied cultures.

Education should instil in the youth of Kenya an understanding of past and present cultures and their valid place in contemporary society. Children should be able to blend the best of traditional values with the changing requirements that must follow rapid development in order to build a stable and modern society.

7. Promote international consciousness and foster positive attitudes towards other nations.

Kenya is part of the international community. It is part of the complicated and interdependent network of peoples and nations. Education should therefore lead the youth of the country to accept membership of this international community with all the obligations and responsibilities, rights and benefits that this membership entails.

8. Promote positive attitudes towards good health and environmental protection.

Education should inculcate in young people the value of good health in order for them to avoid indulging in activities that will lead to physical or mental ill health. It should foster positive attitudes towards environmental development and conservation. It should lead the youth of Kenya to appreciate the need for a healthy environment.

THE SENIOR SCHOOL IN THE COMPETENCY BASED CURRICULUM

Senior School is the fourth level of Basic Education in the Competency Based Curriculum (CBC) that learners will transition to after the Pre-Primary School, Primary School and Junior School levels. The essence of Senior School is to offer learners a pre-university/ pre-career experience in which the learners have an opportunity to choose pathways and tracks where they shall have demonstrated interest and potential at the earlier levels. Senior School comprises three years of education for learners in the 16-18 years age bracket and lays the foundation for further education and training at the tertiary level and the world of work. In the CBC vision, learners exiting this level are expected to be *engaged, empowered and ethical citizens* ready to participate in the socio-economic development of the nation.

At this level, learners shall take seven (7) learning areas as recommended by the *Presidential Working Party on Educational Reforms* (PWPER, 2023). These shall comprise four core learning areas namely: English, Kiswahili, Community Service Learning (CSL) and Physical Education (PE). The learner will select three learning areas depending on career choice, aptitude, interest and personality and with guidance by the career teacher at Senior School.

LEARNING OUTCOMES FOR SENIOR SCHOOL

By the end of Senior School, the learner should be able to:

1. Communicate effectively and utilise information and communication technology across varied contexts.
2. Apply mathematical, logical and critical thinking skills for problem solving.
3. Apply basic research and scientific skills to manipulate the environment and solve problems.
- 4.

- Exploit individual talents for leisure, self-fulfilment, career growth, further education and training. 5.
Uphold national, moral and religious values and apply them in day-to-day life.
6. Apply and promote health care strategies in day-to-day life.
7. Protect, preserve and improve the environment for sustainability.
8. Demonstrate active local and global citizenship for harmonious co-existence.
9. Demonstrate appreciation of diversity in people and cultures.
10. Manage pertinent and contemporary issues responsibly.

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LIST OF SUBJECTS AT SENIOR SCHOOL

Core Subjects	Arts & Sports Science	Social Sciences	Science, Technology, Engineering & Mathematics (STEM)
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1. English 2. Kiswahili/KSL 3. Community Service Learning (CSL) 4. Physical Education (PE)	5. Sports and Recreation 6. Music and Dance 7. Theatre and Film 8. Fine Arts	9. Literature in English 10. Indigenous Languages 11. Fasihi ya Kiswahili 12. Sign Language 13. Arabic 14. French 15. German 16. Mandarin Chinese 17. Christian Religious Education 18. Islamic Religious Education 19. Hindu Religious Education 20. Business Studies 21. History and Citizenship 22. Geography	23. Mathematics 24. Biology 25. Chemistry 26. Physics 27. General Science 28. Agriculture 29. Computer Studies 30. Home Science 31. Aviation 32. Building Construction 33. Electricity 34. Metalwork 35. Power Mechanics 36. Woodwork 37. Media Technology 38. Marine and Fisheries Technology
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LESSON DISTRIBUTION AT SENIOR SCHOOL

Subjects	No of Lessons Per Week (40 minutes per lesson)
Core Subjects 4	

1. English	6
2. Kiswahili	6
3. Physical Education (PE)	3
4. Community Service Learning (CSL)	3
Elective Subjects	
5. Elective 1	6
6. Elective 2	6
7. Elective 3	6
* <i>ICT Skills</i>	2
* <i>PPI</i>	1
* <i>Learner Personal /Group Study</i>	1
Total Number of Lessons	40

***Note:**

a) ICT skills will be offered to all learners to facilitate learning and enjoyment. These skills include accessing, creating,

digital citizenship, and caring for ICT devices. The teacher should use ICT skills for delivery of lessons. b) Every school shall offer Pastoral Programme Instruction (PPI) for moral, spiritual and character development of the learners.

ESSENCE STATEMENT

Chemistry is essential for comprehending the composition, properties of matter and the changes it undergoes. It is a central science that serves as the basis for competencies in science and technical fields essential for industrial and economic growth. This aligns with the African Agenda 2063's and the Kenya Vision 2030's emphasis on Science, Technology and Innovation (STI). Sessional Paper 1 of 2019 also highlights the importance of sustainable education with a focus on STI. The Chemistry curriculum emphasises practical learning and scientific inquiry, employing instructional methods such as inquiry-based learning, project-based learning, and problem-based learning. The specific learning outcomes guide content depth and breadth for cognitive, psychomotor and affective domains in the learner.

The curriculum promotes the use of IUPAC nomenclature, terminologies and equations. The projects integrate chemistry concepts with the environment, fostering creativity and critical thinking through connections to industrial processes and everyday life experiences. Learners are encouraged to initiate their own projects based on the scientific principles learnt. This should create interest, curiosity and fun in the learning of Chemistry. The acquired concepts form the foundation for further learning and training in medicine, engineering, agriculture, aviation and related careers in science and technical fields.

GENERAL LEARNING OUTCOMES

By the end of Senior School, the learner should be able to:

1. Acquire scientific knowledge, skills, values and attitudes for everyday use, and further education and training;
2. Conduct a range of scientific investigations, including collection of qualitative and quantitative data, analysis and interpretation of results;
3. Use scientific language in the study of Chemistry;

4. Develop skills and logical thinking in carrying out project work, field work and excursions, where chemistry can be appreciated on a larger scale;
5. Relate Chemistry in the classroom to modern industrial processes and real-world situations;

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6. Develop critical thinking skills to explain phenomena and address challenges in day-to-day life; 7. Explore, manipulate, manage and conserve the environment for learning and sustainable development; 8. Apply principles of Chemistry and acquired skills to invent, formulate, construct, produce or manufacture useful products from available resources for sustainable development;
9. Develop interest in and appreciation of Chemistry as a subject and its usefulness in society.

SUMMARY OF STRANDS AND SUB-STRANDS

Strand	Sub-Strand	Suggested Number of Lessons
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1.0 Inorganic Chemistry	1.1. Introduction to Chemistry	10
		20
	1.2. The Atom	30
	1.3. The periodic Table	26
	1.4. Chemical Bonding	34
	1.5. Periodicity	
2.0 Physical Chemistry	2.1. Acids and Bases	30
	2.2. Introduction to Salts	30
3.0 Organic Chemistry		
Total Number of lessons		180

Note: The suggested number of lessons per Sub-Strand may be less or more depending on the context.

STRAND 1.0: INORGANIC CHEMISTRY

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Inorganic Chemistry	1.1 Introduction to Chemistry (10 lessons) • Definition of chemistry • Branches of chemistry • Careers in chemistry • Chemistry in our daily lives <i>(mention agriculture, pharmaceutical industry, medicine, manufacturing industry, food industry, entertainment industry, nuclear chemistry, sports industry,</i>	By the end of the sub strand, the learner should be able to: a) explain the meaning of Chemistry as a field of science, b) explore the role of Chemistry in day-to-day life, c) examine the effects of drug and substance use in day-to-day life, d) advocate for a safe and healthy learning environment.	The learner is guided to: • discuss with peers the meaning of Chemistry as a field of science, • discuss with peers the branches of Chemistry, • brainstorm the importance of Chemistry in our daily lives, • search for information using electronic and/or print media on the career opportunities related to Chemistry and how gender stereotyping influences career choices, • discuss with peers the meaning of	How is the study of Chemistry important in our society?

	<i>energy among others).</i>		drug,	
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			prescription, dosage and substance use, ● brainstorm on the learner's rights and responsibilities to a safe and healthy learning environment Project ● develop posters to sensitise the peers/community on the risks of drug and substance use.	
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Core competencies to be developed:

- Communication and Collaboration: The learner explores teamwork with peers by contributing to group decision making through recognising and acknowledging the value of others' ideas during discussions.
- Self-efficacy: The learner effectively communicates while presenting the findings on career opportunities related to Chemistry and how gender stereotypes influence career choices in plenary.
- Digital literacy: The learner uses digital technology effectively to search for information on various career opportunities related to Chemistry.
 - Citizenship: The learner demonstrates critical and constructive dialogue while brainstorming with peers on the consumer rights to information on the drug prescription, substance use and knowledge on their side effects for awareness.

Values:

Respect: The learner shows acceptance while appreciating the opinion of each member as they discuss in groups the various branches of Chemistry.

Pertinent and Contemporary Issues (PCIs)

- Life skills (safety and security): brainstorm on the learner's rights and responsibilities to a safe and healthy learning environment
- Health Promotion Issues (drug and substance use): The learner discusses with peers the meaning of drug prescription, dosage, substance use and their effects in day-to-day life.
- Socio-Economic and Environmental issues (consumer protection): The learner discusses with peers the information on samples of containers of manufactured products and explores the labels on them for marks of quality, expiry dates, nutritional information, preservatives and messages of caution for consumer protection awareness.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
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1.0 Inorganic Chemistry	1.2 The Atom (20 lessons) • Structure of the atom <i>(Energy levels and orbitals)</i> • <i>Isotopes</i> • s and p orbitals <i>(consider the first 20 elements)</i>	By the end of the sub strand, the learner should be able to: a) describe the structure of the atom, b) determine the relative atomic mass of elements, c) write the electron arrangement of elements using <i>s</i> and <i>p</i> notation, d) develop interest in the study of structure of the atom.	The learner is guided to: • review with peers the concept of the structure of the atom, atomic number and mass number, • discuss with peers the relationship between atomic number, mass number and number of electrons in an atom, • illustrate the structure of the atom using Dalton, Rutherford's and Bohr models, • brainstorm the meaning of the terms isotopes and relative atomic mass, • calculate the relative atomic mass of elements from isotopic abundances, • discuss the relationship between energy levels and orbitals in an atom,	How are electrons arranged in an atom?
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			● carry out simple activities to illustrate the order of filling the electrons in the orbitals, ● draw the electron arrangement for the first 20 elements using <i>s</i> and <i>p</i> orbitals, ● watch simulation on the Rutherford Gold Foil experiment and discuss with peers.	
Core competencies to be developed: ● Digital literacy: the learner continuously learns to use digital platforms while watching animations on atomic models, isotopes and electron arrangement of atoms ● Creativity and Imagination: the learner experiments with ideas while modelling the structure of the atom using locally available materials ● Learning to learn: the learner reflects on their own work as they practise illustration of electron arrangements for the first 20 elements using <i>s</i> and <i>p</i> orbitals				
Values: ● Social justice: The learner advocates harmonious relationship in groups while discussing the meaning of isotopes and relative atomic mass ● Integrity: The learner shows self-discipline by appropriately using digital devices to watch animations, simulations or videos on electron arrangement of the atom				

Pertinent and Contemporary Issues (PCIs)

- Socio-Economic and Environmental issues (environmental conservation): the learner responsibly uses locally available materials to model the structure of the atom.
- Life skills (self-esteem): the learner gains confidence while presenting findings on the meaning of isotopes and relative atomic mass.
- Socio-Economic and Environmental issues (global citizenship): the learner brainstorms with peers on the contributions of scientists in the development of atomic theory/models.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
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1.0 Inorganic Chemistry	1.3 The Periodic Table (30 lessons) • Development of the periodic table (Groups and periods). • Ion formation (<i>oxidation number, valency, radicals</i>). • Formulae and chemical equations. (<i>mention alkali metals, alkaline earth metals, halogens, noble</i>	By the end of the sub strand, the learner should be able to: a) relate the position of an element in the periodic table to its electron arrangement, b) illustrate ion formation of elements, c) derive the formulae of compounds, d) write balanced equations for chemical reactions, e) appreciate the role of electron arrangement in the development of the periodic table.	The learner is guided to: • brainstorm in groups, on the historical development of the periodic table, • arrange the first 20 elements of the periodic table into groups and periods • identify the chemical families of elements in the periodic table, • discuss with peers the stability of atoms (<i>loss or gain of electrons</i>), • predict the type of ion formed from a given electron arrangement of an atom (<i>cation and anion</i>), • discuss with peers the relationship between valency	Why is the study of the periodic table important?
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	<i>gases and transition elements)</i>		and oxidation number of elements, • write electron arrangement of ions using <i>s</i> and <i>p</i> notation • infer the valency and oxidation numbers from electron arrangement of elements, • discuss elements with variable oxidation numbers, • practise writing formulae of compounds using valencies and oxidation states of elements and radicals, • write balanced chemical equations for simple chemical reactions.	
Core competencies to be developed: • Communication and Collaboration: the learner listens keenly and actively using open questions while discussing with peers the relationship between atomic number, mass number and number of electrons in atoms of elements. • Critical thinking and problem solving: the learner creates new ideas while drawing diagrams of ions of selected elements using dots and/or crosses to illustrate ion formation. • Learning to learn: the learner learns independently by writing balanced equations for chemical reactions.				

- Citizenship: the learner appreciates contributions made by various scientists on the development of atomic structure and the periodic table.

Values:

Peace: The learner shows compassion as they display tolerance while predicting the type of ion formed from the electron arrangement of elements with peers

Pertinent and Contemporary Issues (PCIs)

- Socio-Economic and Environmental Issues (social cohesion): as the learner discusses with peers how atoms acquire stability.
- Life skills (analytical thinking): as the learner predicts the type of ion formed from a given electron arrangement of an atom.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
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1.0 Inorganic Chemistry	1.4 Chemical Bonding (26 lessons) ● valence electron in bonding (octet rule) ● bond type (<i>ionic, covalent, dative covalent, hydrogen bond, Van der Waals and metallic</i>) <i>Formation of lewis diagrams, resulting structures, physical</i>	By the end of the sub strand, the learner should be able to: a) illustrate bond types in elements, molecules and compounds, b) investigate the relationship between bond types and physical properties of elements, molecules and compounds, c) relate bond types and resultant structures to the uses of elements, molecules and compounds, d) appreciate the uses of different substances based on their bond	The learner is guided to: ● review the concept of stability of atoms (<i>gaining and/or losing electrons</i>). ● discuss in groups, the role of valence electrons in bonding (<i>octet/duplet noble gas configuration</i>). ● discuss with peers, different types of chemical bonds (<i>ionic, covalent, dative covalent, hydrogen bond, Van der Waals forces and metallic</i>). ● draw Lewis structures dot (.) and/or cross (x) diagrams to show bonding in selected elements, molecules and compounds (<i>ionic, covalent, dative covalent, hydrogen</i>	How do chemical bonds influence the properties of substances?
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	<i>properties and uses)</i>	types and structures in day-to-day life.	<i>bond, Van der Waals forces and metallic bonding),</i> ● carry out activities to investigate the physical properties of giant ionic, simple molecular, giant atomic/covalent and giant metallic compounds (<i>solubility, thermal and electrical conductivity, melting point, boiling point</i>), ● brainstorm, in groups, the relationship between bond type and physical properties of elements and compounds (<i>diamond, graphite, sodium chloride, aluminium metal, among others</i>), ● explore the uses of elements, molecules and compounds	
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			based on their bond types and structures, • watch animations, simulations or videos on chemical	
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			bonding and resultant structures. Project: Model bonding in selected molecules and compounds. e.g. NaCl, SiO₂, graphite, diamond (use locally available materials)	
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Core competencies to be developed:

- Learning to learn: the learner learns independently while drawing dot (.) and cross (x) diagrams to illustrate bonding in elements and compounds.
- Digital literacy: the learner uses digital technology to effectively accomplish own task while watching animations, simulations, and videos on chemical bonding.
- Creativity and Imagination: The learner experiments with material to develop models to illustrate bonding in selected elements/compounds using locally available materials.
- Critical Thinking and Problem Solving: The learner interprets and infers with peers the relationship between bond types and resulting structures.

Values:

Love: the learner shows care by avoiding to hurt others as they carry out activities to investigate the physical properties of different structures.

Pertinent and Contemporary Issues (PCIs)

- Socio-Economic and Environmental Issues (environmental conservation): The learner uses locally available materials to make models.
- Life skills (creative thinking): The learner models bonding in selected elements/compounds e.g. SiO_2 , NaCl , graphite and diamond.
- Socio-Economic and Environmental issues (social cohesion): The learner discusses with peers different types of chemical bonds.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Inorganic Chemistry	1.5 Periodicity (34 lessons) • Trends in groups I, II, VII, VIII and period 3 • <i>gradation in size of atoms and ions, appearance, ionisation energy, electron affinity, melting and boiling points.</i> • <i>ductility, malleability and electrical conductivity</i> • chemical	By the end of the sub strand, the learner should be able to: a) describe the trends in physical properties of elements of the periodic table; b) investigate the chemical properties of elements in groups of the periodic table; c) describe the trends in properties across a period, d) outline applications of elements of the	The learner is guided to: • discuss in groups, the trends in physical properties of elements in group I, II, VII and VIII, • carry out experiments to investigate the physical properties of group I and II elements, • carry out experiments to investigate chemical properties of group I and II elements (<i>reaction with oxygen, chlorine, cold water, steam and dilute acids</i>),	1. Why is the study of chemical families important? 2. How does the position of an element in the periodic table influence its properties?

	properties of	periodic table, e) appreciate applications of	● collect and test for the gas produced, ● prepare chlorine gas and carry out experiments to investigate the physical	
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	group I and II elements (<i>reaction with oxygen, chlorine, cold water, steam and dilute acids</i>), • physical properties of chlorine, bromine and iodine (<i>appearance, smell, solubility in water and physical states</i>) • chemical properties of chlorine (<i>reaction with water, hydrogen, metals, displacement reactions</i>,	various elements of the periodic table.	properties of chlorine, bromine and iodine (<i>appearance, smell, solubility in water and physical states</i>), • carry out experiments to investigate the chemical properties of chlorine (<i>reaction with water, reaction with metals, displacement reactions and bleaching action</i>), • discuss the trends in physical properties of period three elements (<i>reaction with water, hydrogen, reaction with metals, displacement reactions, reaction with alkalis and bleaching action</i>) • carry out experiments on reactions of period three elements with oxygen, water, chlorine and dilute acids,	
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	<i>reaction</i> ● physical properties of		● search for information using electronic and/or print media	
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	period three elements (<i>atomic size, ionisation energy, electron affinity, electronegativity, melting and boiling points</i>), - chemical properties of period three elements with (oxygen, water, chlorine and dilute acids), - uses of selected elements in groups I, II, VII and VIII		on the uses of selected elements in groups I, II, VII and VIII discuss the use of selected elements.	
Core competencies to be developed: Communication and Collaboration: as the learner discusses in groups, the trends in physical properties of elements in group I, II, VII and VIII.				

- Self-efficacy: the learner confidently communicates the uses of selected elements while making presentations in plenary.
- Learning to learn: The learner independently learns by practising writing balanced chemical equations for reactions involving group I, II and VII elements.
- Digital literacy: The learner uses digital devices appropriately to search for information on the uses of group I, II, VII and VIII elements.

Values:

Unity: The learner appreciates the participation of each member of the group as they carry out experiments to investigate the physical and chemical properties of elements in the chemical families.

Pertinent and Contemporary Issues (PCIs)

- Learner Support Programmes (healthy inter and intrapersonal relationships): as the learner searches with peers for information on the uses of elements in group I, II, VII and VIII.
- Socio-Economic and Environmental Issues (road safety): The learner discusses the use of selected elements in illuminating street lights for safe roads.
- Life skills (assertiveness): The learner confidently presents their findings on the uses of elements in group I, II, VII and VIII of the periodic table.

Suggested Assessment Rubric

Level Indicator	Exceeds expectations	Meets expectations	Approaches expectations	Below expectations
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Ability to explain the importance of chemistry in our day-to-day life	Correctly explains at least four ways chemistry is important in our day-to-day life.	Correctly explains at least three ways chemistry is important in our day to-day life	Correctly explains at least two ways chemistry is important in our day-to-day life	Correctly explains at least one way chemistry is important in our day to-day life
Ability to illustrate the electron arrangement of elements using <i>s</i> and <i>p</i> notation.	Correctly illustrates the electron arrangement of all the 20 elements using <i>s</i> and <i>p</i> notation.	Correctly illustrates electron arrangement of 15-20 elements using <i>s</i> and <i>p</i> notation.	Correctly illustrates the electron arrangement of <u>10-15</u> elements using <i>s</i> and <i>p</i> notation.	Correctly illustrates the electron arrangement of at most 10 elements using <i>s</i> and <i>p</i> notation.
Ability to write balanced equations for chemical reactions.	Correctly writes balanced equations for all chemical reactions	Correctly writes balanced equations for most chemical reactions	Correctly writes balanced equations for some chemical reactions	Correctly writes a few balanced chemical equations with prompts
Ability to investigate the relationship	Correctly and comprehensively investigates the	Correctly investigates the relationship between bond types	Correctly investigates the relationship between some bond types and	Investigates the relationship between bond types and

Level Indicator	Exceeds expectations	Meets expectations	Approaches expectations	Below expectations
between bond types and physical properties of elements	relationship between bond types and physical properties of elements	and physical properties of elements	physical properties of elements leaving out some details	physical properties of elements leaving out most details
Ability to describe the trends in properties in groups and periods	Correctly describes all the trends in properties in groups and periods.	Correctly describes most trends in properties in groups and periods.	Correctly describes some of the trends in properties in groups and periods.	Describes a few trends in properties in groups and periods.

STRAND 2.0: PHYSICAL CHEMISTRY

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)

2.0 Physical Chemistry	2.1 Acids and bases (30 lessons) • <i>Properties of acids and bases</i> • <i>Reactions of acids with metals, carbonates, hydrogen carbonates, metal oxides and hydroxides.</i> • <i>Indicators</i> • <i>Universal indicator and pH scale.</i> • <i>Weak and strong acids and bases; pH scale and electrical conductivity, (use aqueous solutions</i>	By the end of the sub strand, the learner should be able to: a) explain the characteristics of acids and bases in aqueous solutions, b) describe the chemical properties of acids and bases, c) investigate the strength of acids and bases using the acid-base indicator, d) outline the uses of acids and bases in day to day life,	The learner is guided to: ● carry out experiments to demonstrate dissociation of acids and bases in water, ● carry out experiments on chemical properties of acids, ● perform experiments to investigate reactions of acids and bases with metal oxides and hydroxides ● collect and test for gases produced during the experiments, ● conduct experiments to determine strength of acids and bases using acid base indicator,	What common household substances are acidic and/or basic in nature?
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	<i>of; sulphuric (VI) acid, hydrochloric acid, ethanoic acid, sodium hydroxide, sodium carbonate and ammonia of the same concentration to illustrate)</i> • <i>Applications of acids and bases</i>	e) appreciate the uses of acids and bases in day to day activities.	● carry out activities to compare the electrical conductivity of strong and weak acids and bases, ● properly dispose of waste after experiments, ● search for information on the applications of acids and bases, ● observe safety when handling acids and bases.	
Core competencies to be developed: ● Communication and Collaboration: The learner develops teamwork by contributing to group decision making while working with peers as they carry out laboratory experiments and activities. ● Digital literacy: The learner interacts with digital technology as they search for information on the applications of acids and bases				
Values: Responsibility: The learner develops accountability as they take care of apparatus while using them during experiments				

Pertinent and Contemporary Issues (PCIs):

Environmental issues (safety in the laboratory): The learner observes safety when handling acids and bases

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Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
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2.0 Physical Chemistry	2.2 Introduction to Salts (30 lessons) • Types of salts (<i>normal, acidic, basic and double</i>) • solubility of salts (<i>chlorides, carbonates, nitrates and sulphates</i>) • Methods of preparation (<i>direct synthesis, reactions between; acids and metals, acids and bases, acids and carbonates/hydrogen carbonates, precipitation reaction</i>)	By the end of the sub strand, the learner should be able to: a) classify different salts based on their properties. b) prepare salts using appropriate methods in the laboratory, c) describe the behaviour of salts when exposed to air, d) outline applications of salts in day-to-day life, e) appreciate the applications of salts in day-to-day life.	The learner is guided to: • brainstorm and carry out activities, in groups, to establish the meaning of salt (<i>samples of inorganic fertilisers/table salt/commercial salts found at home and in the school laboratory</i>), • discuss with peers and categorise salts, • carry out experiments to determine the solubility of salts in water and classify them as either soluble or insoluble • carry out experiments to prepare salts • brainstorm on learner's rights and responsibilities	1. How do salts behave when exposed to air? 2. How can salts be prepared in the laboratory?
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	●Behaviour of salts in air (<i>hygroscopic, deliquescent and efflorescent salts</i>) ●Uses of salts (<i>Agriculture, food industry, medicine, paper industry, paints industry, glass industry, laundry among others</i>)		to a safe learning environment when preparing salts, ● write balanced chemical equations for reactions involved in the preparation of salts (<i>ionic equations limited to precipitation reactions</i>), ● carry out experiments to investigate the behaviour of different salts when exposed to the atmosphere ● discuss with peers, the applications of salts, ● search for information using electronic and/or print media on the effects of applications of salts (inorganic fertilisers) on environmental sustainability (<i>water</i>	
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			<i>pollution-eutrophication, soil and air pollution),</i>	
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			• discuss with peers mitigation measures to challenges of using inorganic fertilisers for sustainable economy, health and the environment.	
Core competencies to be developed: • Communication and Collaboration: The learner brainstorms and carries out activities with peers to establish the meaning of salt • Learning to learn: The learner develops relationships by sharing what they have learnt while carrying out experiments to prepare salts • Creativity and imagination: The learner makes observations as they carry out experiments to determine solubility of salts in water.				
Values: • Respect: The learner appreciates the opinion of others as they carry out activities to categorise salts as soluble or insoluble • Unity: the learner cooperates with peers as they discuss the applications of salts				

Pertinent and Contemporary Issues (PCIs)

- Socio-Economic and Environmental Issues (waste management): The learner appropriately disposes of laboratory waste after preparing different salts
- Life skills (self-esteem): The learner confidently makes presentations in plenary on the findings about the effects of excessive use of salts on environmental sustainability

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Suggested Assessment Rubric

Level Indicator	Exceeds expectations	Meets expectations	Approaches expectations	Below expectations
Ability to describe the chemical properties of acids and bases.	Correctly and comprehensively describes all the chemical properties of acids and bases.	Correctly describes most of the chemical properties of acids and bases.	Correctly describes some of the chemical properties of acids and bases,	Correctly describes a few of the chemical properties of acids and bases.
Ability to prepare salts using appropriate methods in the laboratory	Correctly and systematically prepares salts using appropriate methods in the laboratory	Correctly prepares salts using appropriate methods in the laboratory	Correctly prepares salts using appropriate methods in the laboratory missing some steps	Correctly prepares salts using appropriate methods in the laboratory missing most steps

APPENDIX: SUGGESTED RESOURCES, ASSESSMENT METHODS AND NON-FORMAL ACTIVITIES

Strands	Sub-Strands	Suggested Resources	Suggested Assessment Methods	Suggested Non-Formal Activities
Inorganic Chemistry	Introduction to Chemistry	• conventional laboratory resources • improvised resources from the environment • digital devices • online resources	• question and answer method, • written exercises • individual performance assessment • observations • project work	• excursions • field work • science clubs and societies • games • science symposia
	The Atom	• conventional laboratory resources • Improvised resources from the environment • digital devices	• question and answer method, • written exercises • individual performance assessment • observations • project work	• excursions • field work • science clubs and societies • games • science symposia
	The Periodic Table			

	Chemical Families		● practicals	
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	Chemical Bonding	● online resources		
	Periodicity			
Physical Chemistry	Acids and Bases Introduction to Salts	● conventional laboratory resources ● Improvised resources from the environment ● Digital devices ● Online resources	● question and answer method, ● written exercises ● individual performance assessment ● project work ● practicals	● excursions ● field work ● science clubs and societies ● games ● science symposia

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